



Hookworms — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which hookworm stage is infective for humans?

EASY

- A. Filariform larva
- B. Rhabditiform larva
- C. Egg
- D. Cysticercus
- E. Adult worm

2. Which hookworm is the predominant species worldwide?

EASY

- A. *Ancylostoma duodenale*
- B. *Ancylostoma caninum*
- C. *Enterobius vermicularis*
- D. *Taenia solium*
- E. *Necator americanus*

3. What is the most common route of hookworm infection?

EASY

- A. Penetration of the skin by filariform larvae
- B. Ingestion of eggs
- C. Inhalation of dust
- D. Undercooked pork
- E. Retro-infection

4. Which drug is the drug of choice for hookworm infection?

EASY

- A. Praziquantel
- B. Niclosamide
- C. Ivermectin
- D. Mebendazole
- E. Metronidazole

5. A barefoot farmer develops intense itching, oedema and erythema between his toes. What is this reaction?

MEDIUM

- A. Creeping eruption
- B. Ground itch at the site of larval penetration
- C. Cutaneous cysticercosis
- D. Impetigo
- E. Contact eczema

6. A patient with hookworm develops low-grade fever, cough and eosinophilia after the skin lesions. Which stage explains this?

MEDIUM

- A. Adult attachment to the jejunum
- B. Egg deposition in the appendix
- C. Larval migration through the lungs
- D. Cyst rupture
- E. Autoinfection

7. A worm with a large oval buccal capsule bearing two pairs of claw-like teeth is seen. Which species?

MEDIUM

- A. *N. americanus*
- B. *E. vermicularis*
- C. *T. saginata*
- D. *A. duodenale*
- E. *A. caninum*

8. A heavy *A. duodenale* infection can cause a daily blood loss of about:

MEDIUM

- A. 0.03 mL
- B. 0.15 mL
- C. 1 mL
- D. 10–20 mL
- E. 100 mL

9. Which mechanism helps prolong bleeding at the attachment site?

MEDIUM

- A. Vitamin K synthesis
- B. Platelet activation
- C. Complement activation
- D. Hookworm anticoagulant factors that increase clotting time
- E. Fibrin deposition

10. A stool sample shows an oval egg (60 × 40 µm) with a thin shell, a clear space and 7-8 blastomeres. Which is the diagnosis?

MEDIUM

- A. Pinworm infection
- B. Taeniasis
- C. Hookworm infection
- D. Cysticercosis
- E. Trichuriasis

11. Which technique detects hookworm larvae by stool culture?

MEDIUM

- A. Harada-Mori
- B. Kato-Katz
- C. Cellophane tape
- D. Sedimentation
- E. Direct saline smear

12. A public-health team wants to estimate how heavy hookworm infection is in a school. Which technique counts eggs in a known amount of stool?

MEDIUM

- A. Harada-Mori culture
- B. Kato-Katz technique
- C. Cellophane tape test
- D. Serology for IgE
- E. Chest X-ray

13. Eggs first appear in stool about how long after skin penetration?

MEDIUM

- A. 24 hours
- B. 1 week
- C. 4–7 weeks
- D. 10 days
- E. 6 months

14. A patient with chronic hookworm has ankle oedema and low albumin. What is the cause?

MEDIUM

- A. Loss of protein into the gut from feeding worms
- B. Nephrotic syndrome
- C. Heart failure
- D. Filarial lymphatic obstruction
- E. Vitamin B₁₂ deficiency

15. Why does hookworm intensity often increase with age in endemic areas?

HARD

- A. Adults eat more soil
- B. Repeated exposure does not confer effective protective immunity
- C. Larvae mature faster in adults
- D. Adults lack antibodies entirely
- E. Adults have more acid

16. A hookworm product cleaves eotaxin. What is the effect on the host response?

HARD

- A. Reduced neutrophil adhesion
- B. Increased IgE
- C. Increased complement
- D. Increased Th1 cells
- E. Reduced eosinophil recruitment to inflammation sites

17. Neutrophil inhibitory factor benefits the parasite by:

HARD

- A. Cleaving IgA
- B. Neutralising superoxide
- C. Inhibiting C1q
- D. Cleaving eotaxin
- E. Blocking neutrophil adhesion to endothelium and H₂O₂ release

18. A child with hookworm has growth retardation and poor digestion. Which parasite product is implicated?

HARD

- A. Elastin-cleaving metalloprotease
- B. Superoxide dismutase
- C. C-type lectin
- D. Neutrophil inhibitory factor
- E. A serine protease inhibitor that interferes with pancreatic-enzyme digestion

19. Which statement about hookworm in pregnancy is correct?

HARD

- A. Trans-placental transmission is the main route
- B. Iron deficiency and malnutrition affect fetal growth, birth weight and maternal survival
- C. It has no maternal effect
- D. Eosinophilia damages the fetus
- E. Only *A. caninum* is dangerous

20. A traveller has an itchy, wandering track in the skin of the foot but no eggs in stool. Which explanation is most consistent?

HARD

- A. *N. americanus* adults in the jejunum
- B. Filariform larvae maturing normally
- C. Dog or cat hookworm larvae that cannot mature in humans
- D. Pinworm eggs
- E. *Taenia cysticerci*

✓ Answer key

Q	Answer	Difficulty	Why
1	A	EASY	Filariform larvae are non-feeding and infective; rhabditiform larvae hatch in soil.
2	E	EASY	<i>N. americanus</i> is the predominant species; <i>A. duodenale</i> is the Old World hookworm.
3	A	EASY	Skin penetration (feet, hands, buttocks) is the most common route.
4	D	EASY	The lecture names mebendazole as the drug of choice, effective against adults and larvae.
5	B	MEDIUM	Ground itch is an allergic reaction where filariform larvae enter the skin.
6	C	MEDIUM	Large numbers of larvae passing through the lungs cause minute haemorrhages, leukocyte infiltration and eosinophilia.
7	D	MEDIUM	<i>A. duodenale</i> has a large, oval capsule with teeth; <i>N. americanus</i> has cutting plates.
8	D	MEDIUM	Each <i>A. duodenale</i> worm loses 0.15–0.26 mL/day; heavy infection totals 10–20 mL/day.
9	D	MEDIUM	Anticoagulant secretions prolong blood seepage from attachment sites.
10	C	MEDIUM	The description matches hookworm eggs; pinworm eggs are planoconvex.
11	A	MEDIUM	Harada-Mori is the stool culture method for larvae; Kato-Katz counts eggs.
12	B	MEDIUM	Kato-Katz estimates infection intensity by egg counting; Harada-Mori cultures larvae.
13	C	MEDIUM	Fertilised females begin laying eggs about 4–7 weeks after infection.
14	A	MEDIUM	Blood and protein loss produce hypoproteinaemia and oedema.
15	B	HARD	The lecture notes the absence of effector immunity explains rising intensity with age.
16	E	HARD	Elastin-cleaving metalloproteases degrade eotaxin and inhibit eosinophil recruitment.
17	E	HARD	NIF acts on neutrophils; IgA cleavage is by <i>N. americanus</i> proteases.
18	E	HARD	The protease inhibitor causes malabsorption and growth retardation.
19	B	HARD	The lecture also says trans-placental transmission is very rare.
20	C	HARD	Creeping eruption: larvae wander in superficial skin and rarely reach the lung.